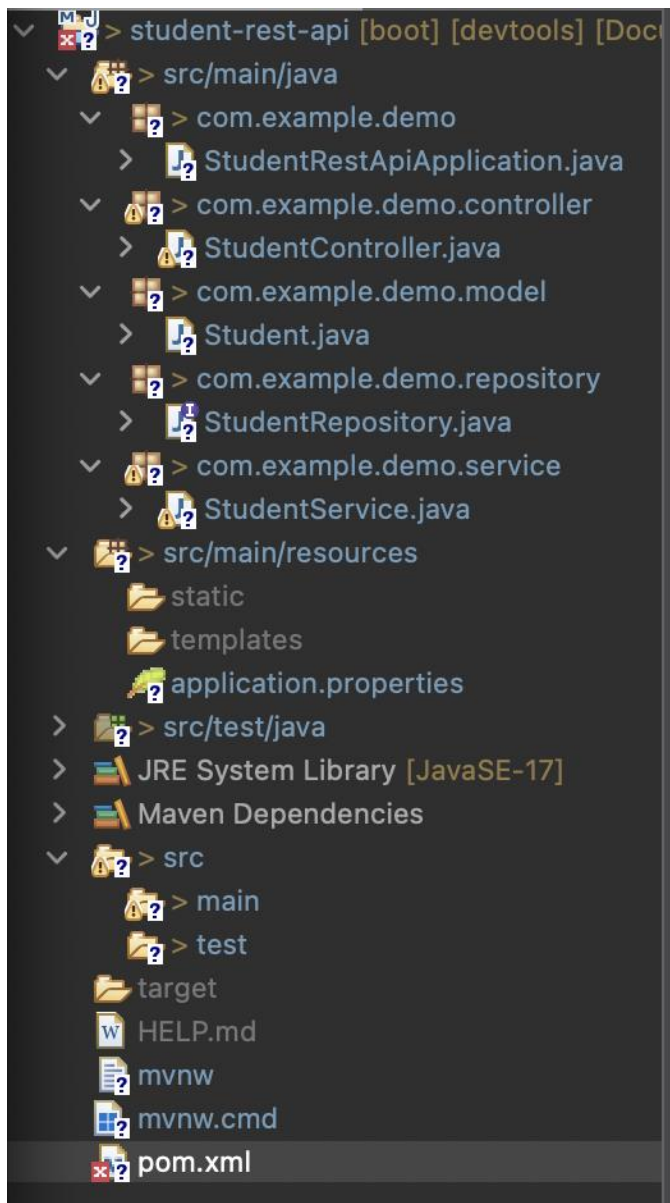


DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT

Practical Exercise 1 – Student Management REST API

- Name :M.Srinikitha
- Roll Number :2520030161
- Section :8

Project Structure:



MySQL Server:

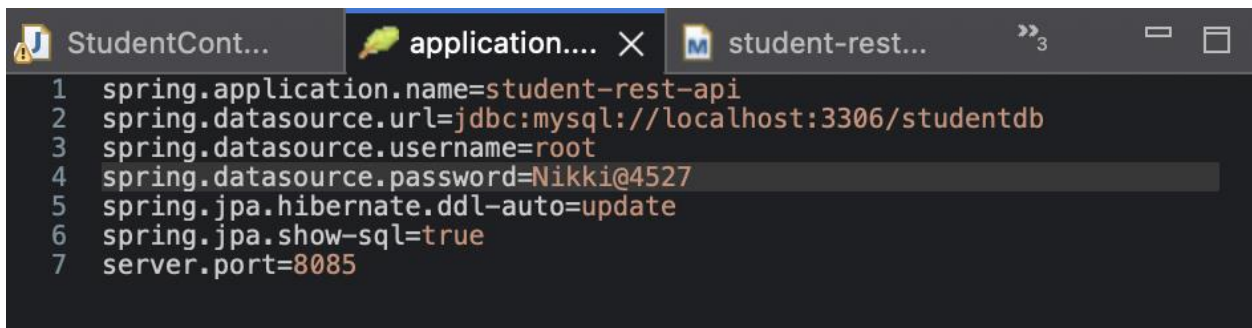
```
[mothenikhitha@Mothes-MacBook-Air ~ % ls /usr/local/mysql/bin/mysql
/usr/local/mysql/bin/mysql
[mothenikhitha@Mothes-MacBook-Air ~ % /usr/local/mysql/bin/mysql -u root -p
[Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 9.7.1 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```
[mysql> CREATE DATABASE studentdb;
Query OK, 1 row affected (0.019 sec)
```



The screenshot shows an IDE with three tabs: 'StudentCont...', 'application.... X', and 'student-rest...'. The 'application.... X' tab is active and displays the following configuration properties:

```
1 spring.application.name=student-rest-api
2 spring.datasource.url=jdbc:mysql://localhost:3306/studentdb
3 spring.datasource.username=root
4 spring.datasource.password=Nikki@4527
5 spring.jpa.hibernate.ddl-auto=update
6 spring.jpa.show-sql=true
7 server.port=8085
```

Postman:

The image shows the Postman application interface. At the top, the request method is set to **POST** and the URL is `http://localhost:8085/students`. A **Send** button is visible. Below the URL bar, the **Body** tab is selected, and the **raw** sub-tab is chosen. The **JSON** format is selected, and the **Schema** and **Beautify** options are visible. The request body is a JSON object:

```
1 {  
2   "name": "Ramya",  
3   "department": "CSE",  
4   "email": "ramya@gmail.com"  
5 }
```

Below the request body, the response status is **200 OK**, with a response time of **19 ms** and a size of **224 B**. The **Body** tab is selected for the response, and the **JSON** format is chosen. The response body is a JSON object:

```
1 {  
2   "id": 3,  
3   "name": "Ramya",  
4   "department": "CSE",  
5   "email": "ramya@gmail.com"  
6 }
```

GET ▼ http://localhost:8085/students Send ▼

Body ▼ Cookies

raw ▼ JSON ▼ Schema Beautify

```
1 {
2   "name": "Ramya",
3   "department": "CSE",
4   "email": "ramya@gmail.com"
5 }
```

Body ▼ 🔄 200 OK • 12 ms • 226 B • 🌐 e.g. ...

{} JSON ▼ ▶ Preview 🖼️ Visualize ▼ 🔄 ≡ 🔍 📄 🔗

```
1 [
2   {
3     "id": 3,
4     "name": "Ramya",
5     "department": "CSE",
6     "email": "ramya@gmail.com"
7   }
8 ]
```

Test it with MySQL Server:

```
[mysql> USE studentdb;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
[mysql> SHOW TABLES;
+-----+
| Tables_in_studentdb |
+-----+
| student              |
+-----+
1 row in set (0.002 sec)
```

```
[mysql> SELECT * FROM student;
+----+-----+-----+-----+
| id | department | email          | name |
+----+-----+-----+-----+
| 1  | CSE        | ramya@gmail.com | Ramya |
+----+-----+-----+-----+
1 row in set (0.001 sec)
```

Testing with cURL:

```
[mothenikhitha@Mothes-MacBook-Air ~ % curl -X POST http://localhost:8085/students
-H "Content-Type: application/json" -d '{"name":"Ramya","department":"CSE","email":"ramya@gmail.com"}'
[{"id":2,"name":"Ramya","department":"CSE","email":"ramya@gmail.com"}]
mothenikhitha@Mothes-MacBook-Air ~ % curl http://localhost:8085/students
[{"id":2,"name":"Ramya","department":"CSE","email":"ramya@gmail.com"}]
mothenikhitha@Mothes-MacBook-Air ~ % curl -X PUT http://localhost:8085/students/
2 -H "Content-Type: application/json" -d '{"name":"Ramya Updated","department":"
ECE","email":"ramya_updated@gmail.com"}'
[{"id":2,"name":"Ramya Updated","department":"ECE","email":"ramya_updated@gmail.c
om"}]
mothenikhitha@Mothes-MacBook-Air ~ % curl -X DELETE http://localhost:8085/studen
ts/2
[Deleted Successfully]
mothenikhitha@Mothes-MacBook-Air ~ % curl http://localhost:8085/students
[]
mothenikhitha@Mothes-MacBook-Air ~ %
```